📖 Heavy Load Operation Adjustment

8. Load operating test
    - B1. Low-frequency loading performance is below 1/10 of rated frequency:
        - a. If the low-frequency loading performance is insufficient, or the rotor speed is not smooth, increase Pr.10-31 (Current command of I/F mode).
        - b. If the low-frequency current is large, decrease Pr.10-31 (Current command of I/F mode).
    - B2. Test the with-load accelerating performance:
      When the motor operates in 1/10 of rotor speed and above, if the speed cannot follow the acceleration time during accelerating, or the current stalls, increase Pr.07-38 (PMSVC voltage feedback forward gain).
9. Stability test at constant speed operation: if the motor operates stably at constant speed
    - a. If the motor output rotor speed presents periodic low-frequency wave, increase Pr.10-34 (PM sensorless speed estimator low-pass filter gain), or increase Pr.10-32 (PM FOC sensorless speed estimator bandwidth).
    - b. If the output frequency reflects high frequency vibration, decrease Pr.10-34 or decrease Pr.10-32.

- PMSVC Related Parameters

  Refer to Section 12-1 Description of Parameter Settings for more details.

Parameter	Description	Unit	Default	Setting Range
Pr.07-24	Torque command filter time	sec.	0.5	0.001–10
Pr.07-26	Torque compensation gain	N/A	0	0–5000
Pr.07-38	PMSVC voltage feedback forward gain	N/A	1.0	0.5–2.0
Pr.10-31	I/F mode, current command	%	40	0–150
Pr.10-32	PM FOC sensorless speed estimator bandwidth	Hz	5.00	0.00–600.00
Pr.10-34	PM sensorless speed estimator low-pass filter gain	N/A	1.00	0.00–655.35
Pr.10-39	Frequency point to switch from I/F mode to PM sensorless mode	Hz	20.00	0.00–599.00
Pr.10-40	Frequency point to switch from PM sensorless mode to V/F mode	Hz	20.00	0.00–599.00
Initial Angle Estimating Parameters				
Pr.10-42	Initial angle detection pulse value	N/A	1.0	0.0–3.0
Pr.10-51	Injection frequency	Hz	500	0–1200
Pr.10-52	Injection magnitude	V	15.0 / 30.0	0.0–200.0
Pr.10-53	PM initial rotor position detection method 0: Disable 1: Using I/F current command (Pr.10-31) to attract the rotor to zero degrees 2: High frequency injection 3: Pulse injection	N/A	0	0–3